Operational Context & Baseline Assumptions

SYNCOS ORCHESTRATION ENGINE

Before analyzing the generated telemetry data to identify throughput losses, we establish a strict set of baseline assumptions regarding the manufacturing site's operations, fleet composition, and the existing software behavior of the orchestration layer.

Explicit Assumptions

- **Operational & Fleet Context**
  The facility operates continuously 24/7 across three standard production shifts.
- **Fleet Composition**
  The mixed fleet consists exactly of 10 robots: 5 autonomous forklifts handling pallets and 5 dual-conveyor robots handling intermediate materials.
- **System Baseline**
  SyncOS already possesses basic payload compatibility awareness, ensuring tasks are only assigned to the correct vehicle class.
- **The Orchestration Assumption (FIFO)**
  We assume SyncOS currently utilizes a strict First-In, First-Out (FIFO) logic, assigning tasks purely in the chronological order they arrive.

METHODOLOGY CONSTRAINT

By strictly isolating these assumptions, the upcoming data analysis must rely entirely on the telemetry records to prove the existence of physical site bottlenecks or validate the feasibility of a 10% throughput increase. No external conclusions have been pre-assumed beyond this operational baseline.